GEOMETRY / MASS / DYNAMICS / TEST ENVELOPE

Prototype P0-V1 Technical & Performance Specification

VTX-P0-V1 / P0-V1 / ISSUE A / 30 OCT 2026

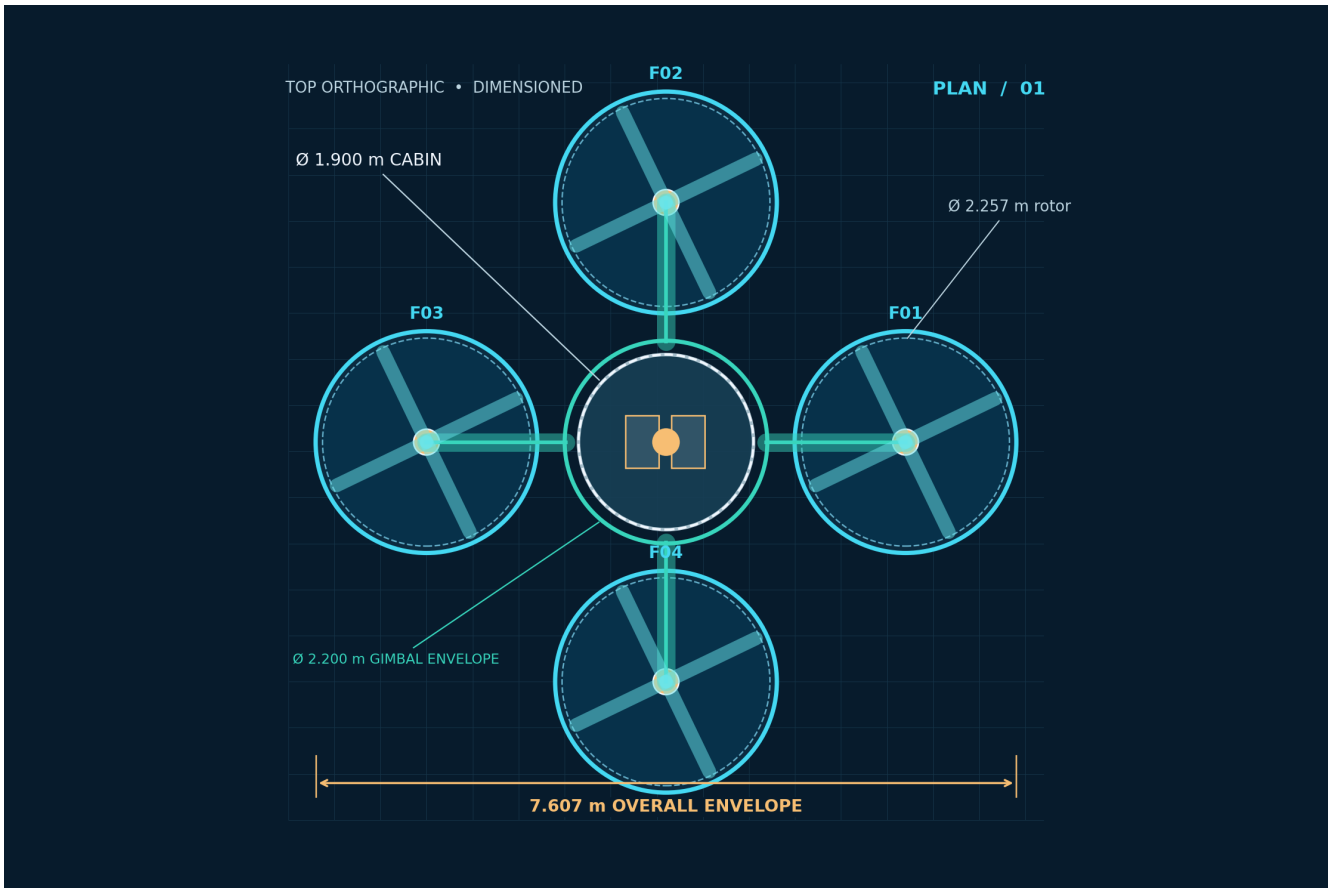
ENGINEERING CONCEPT / NOT A RELEASED DESIGN: geometry is an internally consistent concept baseline; component masses, aerodynamic coefficients, engine-specific fuel consumption, crashworthiness and flight-control performance are not substantiated by qualification tests.

Control definition: P0-V1 is a single-turbine, mechanically distributed, four-ducted-rotor VTOL with a two-axis gimballed two-seat spherical cabin. It is a controlled analytical prototype concept, not a certified, validated or flight-proven aircraft.

01 / Configuration register

PARAMETER	P0-V1 CONTROLLED VALUE
Cabin outer diameter	1.900 m (radius 0.950 m)
Cabin + gimbal outer envelope	2.200 m (radius 1.100 m)
Rotor count / arrangement	Four identical ducted fans, centres on cardinal radial axes
Rotor disk diameter	2.2568 m (each)
Fan outer duct diameter	2.4068 m (each; +75 mm outer radial allowance)
Total rotor disk area	16.000 m²
Fan hub radius from centre	2.600 m
Overall dimension-controlled span	7.6068 m
Gimbal-to-duct radial clearance	296.6 mm in top view (static)
Stated model air density	1.225 kg/m³ (sea level benchmark)
Vertical envelope and landing-gear height	TBD - not dimension-controlled
3D centre-of-gravity/inertia properties	TBD - CAD model and mass properties required

Schematic geometry



Rotor tip clearance and the radial 296.6 mm gap are different dimensions. Dynamic deflection, blade growth, cabin gimbal travel and rotor-to-cabin clearances have not yet been verified.

02 / Mass, materials and component identification

ID	SUBSYSTEM	MASS KG
P01	Turboshaft installation	132
P02	Four ducted fan modules	119
P03	Four-way transmission	72
P04	Spherical cabin module	88
P05	Two-axis cabin gimbal	35
P06	Main structure	75
P07	Fuel installation	29
P08	Flight-control avionics	23
P09	Two seat/restraint systems	25
P10	Landing gear	24
P11	Electrical and cooling	49
-	TOTAL EMPTY MASS	671

OPERATING CONDITION	MASS KG	STATUS
Empty design target	671	Sum of P01-P11 allocations, not weighed
Occupant allowance	170	Two persons combined
Short test fuel	60	Jet A-1
Short mission takeoff mass	901	Nominal model case
Alternative 110 kg fuel loading	951	Sensitivity only; check actual volume and tank structure

Design materials and assemblies

Cabin: 35% nominal glazed surface area, laminated aviation glazing allowance, CFRP sandwich opaque structure and qualified flame-resistant internal trim. Main frame: carbon-composite load ring and stiffened box supports; metallic inserts at highly concentrated load paths. Fan modules: composite blades/duct plus metallic bearing and gearing elements. These statements identify candidates, not released laminate schedules.

03 / Propulsion and power distribution

PARAMETER	VALUE / REQUIREMENT
Prime mover	Rolls-Royce M250-C30P reference (not procured)
Reference takeoff rating	484.7 kW (650 shp; reference engine takeoff rating)
Installed max-continuous placeholder	469.8 kW (630 shp reference; installed rating not validated)
Rotor reference design speed	800 rpm study point, NOT released blade speed
Rotor tip speed at study rpm	94.5 m/s
4-blade passing frequency at study rpm	53.3 Hz
Transmission functional topology	Single power source -> four-way gearbox -> four shaft branches; differential thrust via four independent collective-pitch hubs
Approx gearbox ratio if engine N2 is 6016 rpm	7.52:1 to reach 800 rpm; gearing not designed
Auxiliary distribution	Sensors, gimbal actuators, four pitch servo paths and independent flight ECU on dedicated bus
Fuel type	Jet A-1; compatible SAF only after turbine approval and property assessment

Reference Rolls-Royce C30P data are from [S01]. The engine output-speed 6016 rpm and dry weight approximately 245 lb are reference publication values, not an installed P0 component weighing or guarantee. Installed driveshaft, gearbox and rotor limits constrain available rotor power.

04 / Formulae and numerical flight model

Primary hover-equivalent induced shaft power (actuator-disk approximation):

$P_{hover} = (m\ g / \eta) \times \sqrt{[(m\ g) / (2\ \rho\ A)]}$

Turn reference: 27.0 km at 100.0 km/h (972 s) is modelled as an equivalent continuous-radius turn, plus 80 s of hover/transition power; total 1052 s, 17 minutes 32 seconds. This is a test-energy benchmark rather than a complete time-resolved control simulation.

Drag force: $D = 0.5\ \rho\ V^2\ C_D\ A$; baseline assumed $C_D A = 1.65\ m^2$. Net turn thrust vector magnitude: $T = \sqrt{[(m\ g)^2 + (m\ V^2/R)^2 + D^2]}$. Forward-flight shaft power is approximated as induced-power-equivalent for T plus $D\ V / \eta_{prop}$. This is not a blade-element-momentum or edgewise duct-performance computation.

MODEL INPUT	NOMINAL
Rotor aerodynamic efficiency η	0.72 (unverified aggregate)
Forward propulsive efficiency η_prop	0.75 (unverified)
Engine specific fuel consumption	0.400 kg/kWh (unverified part-load placeholder)
Fuel density	0.800 kg/L (planning assumption)
Fuel pump price	GBP 1.45/L, Brighton September 2026 [S10]
Main turn reference radius	500 m; stress-turn radius 200 m

05 / Numerical mission results

CASE	HOVER kW	TURN kW	E kWh	FUEL kg	GBP
P0 / 500 m radius, 901 kg	184.3	217.7	62.88	25.15	45.58
P0 / 200 m radius, 901 kg	184.3	235.3	67.61	27.05	49.02
P0 / 500 m, full fuel 951 kg	199.9	233.5	67.49	27.00	48.93
P0 / pessimistic envelope	221.2	282.0	81.07	40.53	73.47

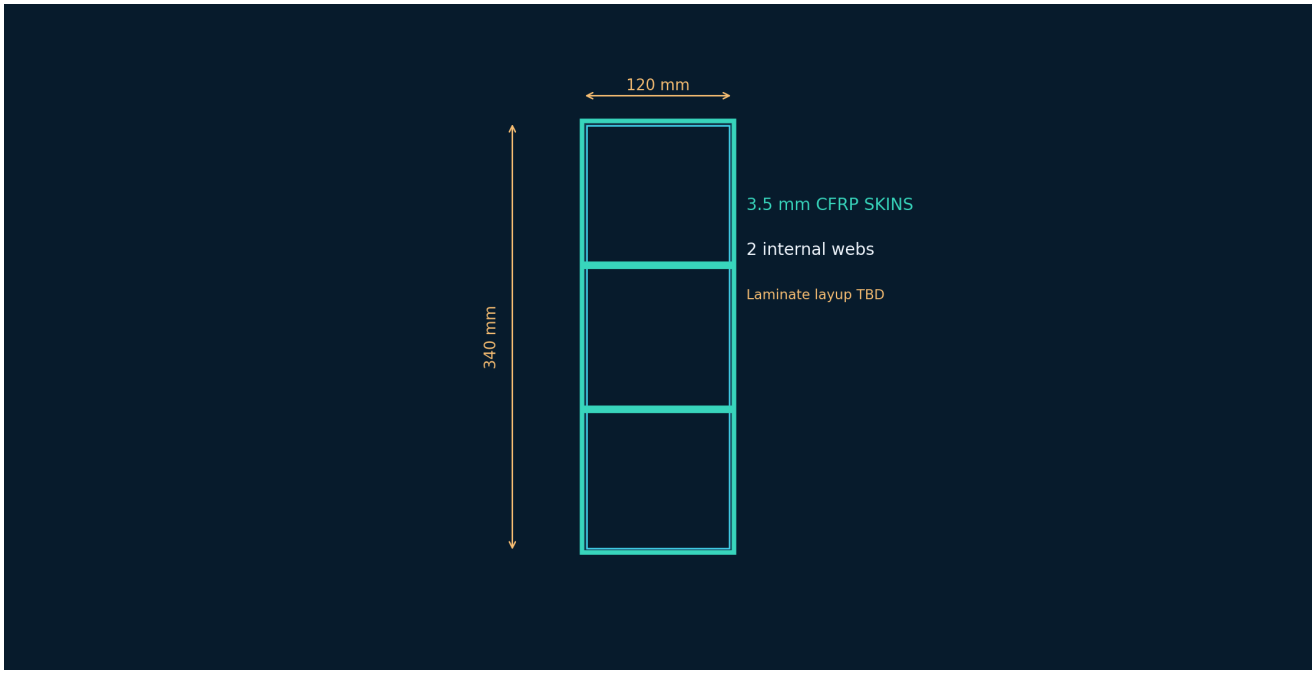
Pessimistic case simultaneously assumes $\eta = 0.60$, $C_D A = 3.0\ m^2$ and SFC 0.50 kg/kWh. Nominal results use constant starting mass, therefore do not claim fuel-burn feedback, measured thrust maps, thermal derate or terrain effects. All quoted mission energy is shaft energy, not fuel chemical energy.

06 / Loads and design constraints

TEST OR LOAD CASE	ESTIMATED QUANTITY	RELEASE STATUS
Static hover at 901 kg	8,839 N total; 2,210 N/rotor	Actuator-disk estimate
1.5g collective thrust	3,315 N/rotor	Structural screening only
Hover shaft power / 901 kg	184.3 kW	Analytical
1.5g induced shaft power	338.7 kW	Power ∝ thrust^1.5
1.5g power plus estimated drag	367.5 kW	Not full transient power model
500 m-radius turn	0.157 g lateral; 8.94° nominal bank	Analytical
200 m-radius turn	0.393 g lateral; 21.47° nominal bank	Analytical
Sample gimbal occupied-mass 1.5g load	~4.2 kN at 283 kg example occupied capsule	Bearing moment arm / allowable TBD
2 m/s landing over 0.20 m stroke	Mean 2.02g reaction equivalent, approximate	Peak landing loads not determined
3 m/s landing over 0.20 m stroke	Mean 3.29g reaction equivalent, approximate	Ground / gear test required

Composite arm reference section

Illustrative 1.50 m cantilever from ring to rotor hub; 340 × 120 mm CFRP box section, 3.5 mm skins, two internal webs. At illustrative 1.5g thrust and 900 Nm torque the simplified beam study predicted ~1.4 mm tip deflection, ~0.33° twist and ~43.9 Hz bending mode. These values are NOT structural allowables; joint compliance, laminate coupling, web local buckling and load concentrations were excluded.



07 / Flight control and gimballed cabin

FUNCTION	CONCEPT CONTROL / SENSOR PATH
Vehicle attitude	IMU feedback commands four independent collective-pitch actuators on common-speed shaft-driven fans; vehicle roll/pitch/yaw
Translational manoeuvring	Controlled vehicle tilt and thrust magnitude; fixed planar rotors cannot independently vector thrust
Cabin coordination	Two-axis pitch/roll gimbal; bank to resultant acceleration in turns rather than always remain horizontal
Occupant asymmetry	Measured/estimated CG offset, preflight trim and bounded control corrections
Sensor suite	Independent IMUs, rotor speed/torque measurement, air data, fuel, vibration, bus power, gimbal position
Power interruption	Engine-out and driveline faults are unresolved critical hazards; first programme uses uncrewed tethered testing
Rotor-out allocation	Four nonnegative variable-pitch lift channels cannot generally preserve all 4 thrust/moment DOF after one fan or drive branch fails

Failure-mode register / assessment

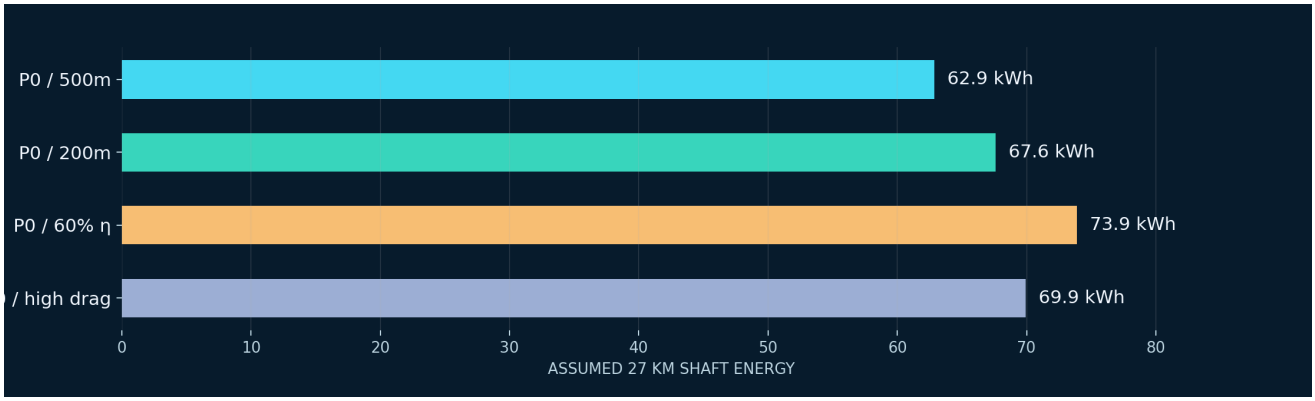
ID	FAILURE MODE	STATUS	MITIGATION REQUIREMENT
FM01	Rotor / shaft branch loss	CONTROL AUTHORITY: FAIL / not substantiated	Fail-to-hold control allocation possible; system-level redundancy needed.
FM02	Single engine failure	CRITICAL / unmitigated in P0	Common prime mover removes all nominal lift power.
FM03	Duct / blade damage	OPEN	Containment and stiffness against rotor-out deformation unverified.
FM04	Cabin bearing / gimbal jam	OPEN	Loads may transmit directly to occupants and disturb controlled cabin attitude.
FM05	Composite joint debond / buckling	OPEN	Material data and finite-element model required; local panel buckling risk.
FM06	Inlet distortion / crosswind	OPEN	Forward-flight fan maps and yaw side forces not yet measured.
FM07	Asymmetric occupants/fuel CG	OPEN	CG and inertia model not complete; trim/saturation study required.
FM08	Rotor modal interference	OPEN	800 rpm / 4-blade 53.3 Hz is an illustrative excitation, not rotor validated.

08 / Test definition and evidence gates

VERIFICATION DOMAIN	ACCEPTANCE EVIDENCE	CURRENT STATUS
Geometry	Full CAD envelopes and verified angular clearance, worst-case gimbal travel	NOT COMPLETE
Mass	Weighed parts / CG and complete moments of inertia	NOT COMPLETE
Structure	Orthotropic FEA, fatigue, joint coupon data, load proof and drop test	NOT COMPLETE
Aerodynamics	Measured isolated fan plus full-vehicle CFD / tunnel maps	NOT COMPLETE
Propulsion	Installed gearbox output, temperature, fuel burn and transient map	NOT COMPLETE
Control	SIL/HIL off-nominal simulations, saturations and actuator latency	NOT COMPLETE
Ground test	Rotor guard, containment, vibration, cooling, telemetry, abort logic	NOT COMPLETE
Flight	Tethered uncrewed hover followed by supervised restricted-range manoeuvres	NOT STARTED

Programme hypothesis to test

The design may complete a 27 km 100 km/h equivalent mission for approximately 25.2 kg Jet A-1 on the nominal low-order model, with adequate nominal turbine takeoff power. This is a feasibility hypothesis only. Before a human-occupied or untethered prototype can be justified, the single-engine and loss-of-one-fan control risks must be resolved or the test campaign redesigned to contain their consequences.



09 / Standards, research and source references

S01 / Rolls-Royce. M250/RR300 FIRST Network Directory (2024); C30P 650 shp takeoff, nominal 245 lb dry, published SFC table. OPEN SOURCE

S02 / Bell / 206L-4. Engine takeoff and maximum-continuous engine ratings vs installed transmission limitations (2005 product specification). OPEN SOURCE

S03 / Hine et al., 2025. Design and Analysis of a Liftfan for eVTOL Aircraft. Experimental rotor/stator optimisation and nacelle forward-flight losses. DOI:10.1115/1.4067673. OPEN SOURCE

S04 / Hu et al., 2021. Hovering efficiency optimization of the ducted propeller with weight penalty taken into account. DOI:10.1016/j.ast.2021.106937. OPEN SOURCE

S05 / Akturk & Camci, 2014. Tip-clearance treatment experiments in ducted VTOL fans. DOI:10.1115/1.4023469. OPEN SOURCE

S06 / Critchfield & Ning, 2026. Mission-focused multidisciplinary eVTOL propulsion optimisation. DOI:10.2514/1.C038445. OPEN SOURCE

S07 / Li, Chen & Jia, 2023. Experimental investigation on hover performance of a ducted coaxial-rotor UAV; rotor location and clearance influence. OPEN SOURCE

S08 / Liu et al., 2024. Impact response and crashworthy design of composite fuselage structures: An overview. OPEN SOURCE

S09 / Muniraj & Sreehari, 2021. Experimental damage evaluation of honeycomb sandwich with composite face sheets. OPEN SOURCE

S10 / Brighton City Airport. Jet A-1 gross indicative pump price GBP 1.45/L from 5 September 2026. OPEN SOURCE

S11 / Shobdon Airfield. Jet A-1 quoted GBP 1.82/L nonmember, 2 October 2026; regional price variation. OPEN SOURCE

Design maturity: first engineering prototype specification, Issue A. Not a fabrication drawing set, stress report, aircraft type design, flight clearance or certification claim.